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In [19]: # Install required libraries (uncomment if needed)
# !pip install pandas numpy matplotlib scikit-learn

# Import libraries with standard aliases
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets
import warnings
warnings.filterwarnings('ignore') # Hide warning messages for cleaner
output

print("✅ All libraries imported successfully!")
print(f"Pandas version: {pd.__version__}")
print(f"NumPy version: {np.__version__}")

# Load a simple dataset (Iris flowers - a classic beginner dataset)
from sklearn.datasets import load_iris

# Load the data
iris = load_iris()
print("Dataset loaded successfully!")
print(f"Dataset shape: {iris.data.shape}")
print(f"Features: {iris.feature_names}")
print(f"Target classes: {iris.target_names}")

# Convert to pandas DataFrame for easier handling
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target_names[iris.target]

# Display first few rows
print("First 5 rows of our dataset:")
print(df.head())

print("\nDataset info:")
print(df.info())

# Create a simple scatter plot
plt.figure(figsize=(10, 6))

# Plot sepal length vs sepal width, colored by species
species_colors = {'setosa': 'red', 'versicolor': 'blue', 'virginica':
'green'}

for species in df['species'].unique():
    species_data = df[df['species'] == species]
    plt.scatter(species_data['sepal length (cm)'],
                species_data['sepal width (cm)'],
                c=species_colors[species],
                label=species,
                alpha=0.7)
```

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plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')
plt.title('Iris Dataset: Sepal Length vs Sepal Width')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()

# print("🎉 Congratulations! You've created your first data
visualization!")

# Basic statistical analysis
print("Basic Statistics for Iris Dataset:")
print("-" * 40)

# Calculate mean values for each species
species_means = df.groupby('species').mean()
print("\nMean values by species:")
print(species_means)

# Count samples per species
species_counts = df['species'].value_counts()
print("\nSamples per species:")
print(species_counts)

# Task 1: Create a simple calculation using NumPy
# Calculate the mean and standard deviation of sepal length

sepal_lengths = df['sepal length (cm)']

# Your code here:
mean_sepal_length = np.mean(sepal_lengths)
std_sepal_length = np.std(sepal_lengths)

print(f"Mean sepal length: {mean_sepal_length:.2f} cm")
print(f"Standard deviation: {std_sepal_length:.2f} cm")

# Verification (don't modify)
assert isinstance(mean_sepal_length, (float, np.floating)), "Mean should be
a number"
assert isinstance(std_sepal_length, (float, np.floating)), "Std should be a
number"
print("✅ Task 1 completed successfully!")

# Task 2: Create a simple bar chart showing species counts
species_counts = df['species'].value_counts()

plt.figure(figsize=(8, 6))
plt.bar(species_counts.index, species_counts.values, color=['red', 'blue',
'green'])
plt.title('Number of Samples per Species')
plt.xlabel('Species')
plt.ylabel('Count')
```

```
plt.show()

print(f"Species distribution: {dict(species_counts)}")
print("✅ Task 2 completed successfully!")
```

✅ All libraries imported successfully!

Pandas version: 2.2.3

NumPy version: 2.1.3

Dataset loaded successfully!

Dataset shape: (150, 4)

Features: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']

Target classes: ['setosa' 'versicolor' 'virginica']

First 5 rows of our dataset:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

	species
0	setosa
1	setosa
2	setosa
3	setosa
4	setosa

Dataset info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 150 entries, 0 to 149

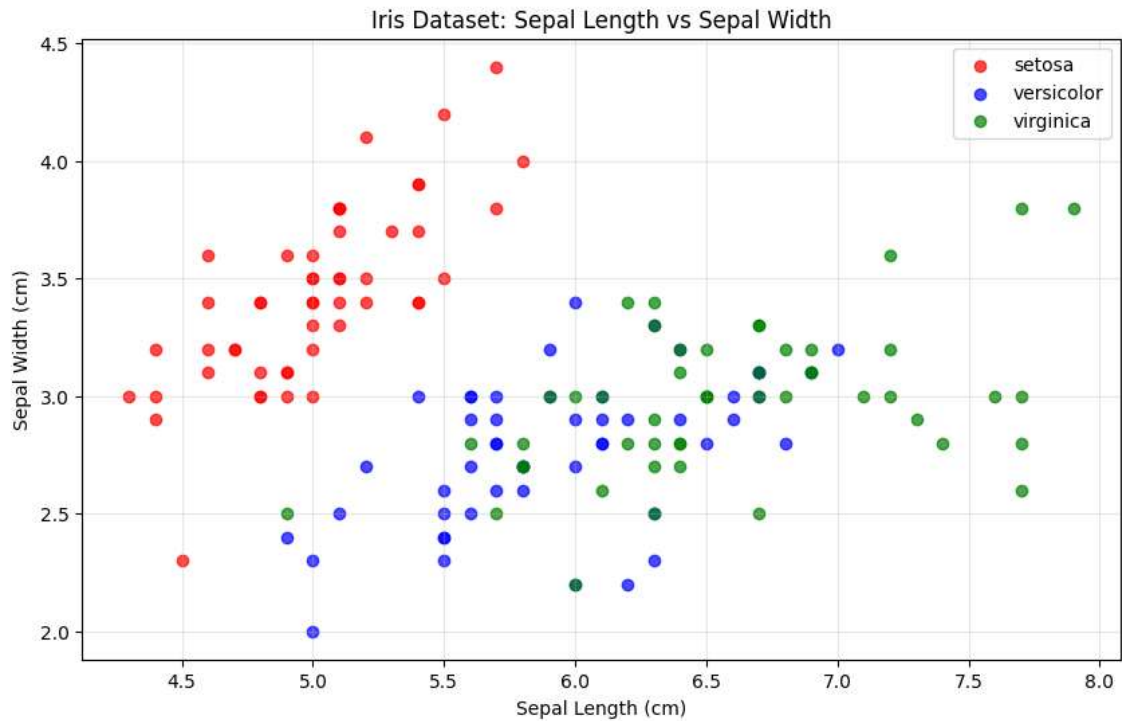
Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	sepal length (cm)	150 non-null	float64
1	sepal width (cm)	150 non-null	float64
2	petal length (cm)	150 non-null	float64
3	petal width (cm)	150 non-null	float64
4	species	150 non-null	object

dtypes: float64(4), object(1)

memory usage: 6.0+ KB

None



Basic Statistics for Iris Dataset:

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Mean values by species:

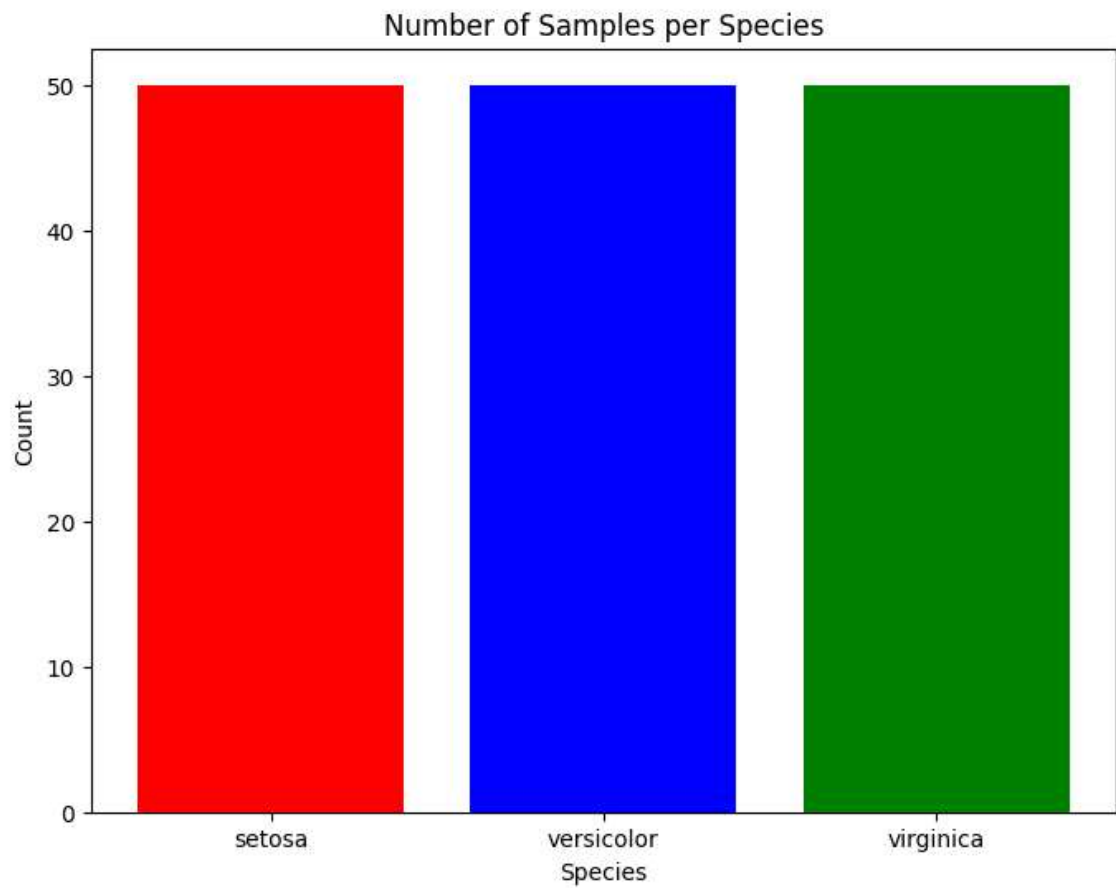
	sepal length (cm)	sepal width (cm)	petal length (cm) \
species			
setosa	5.006	3.428	1.462
versicolor	5.936	2.770	4.260
virginica	6.588	2.974	5.552

	petal width (cm)
species	
setosa	0.246
versicolor	1.326
virginica	2.026

Samples per species:

species	
setosa	50
versicolor	50
virginica	50

Name: count, dtype: int64
Mean sepal length: 5.84 cm
Standard deviation: 0.83 cm
✅ Task 1 completed successfully!



Species distribution: {'setosa': np.int64(50), 'versicolor': np.int64(50), 'virginica': np.int64(50)}

✅ Task 2 completed successfully!